Optimal Convergecast TDMA Frame Length in Multi-Hop Radio Chains and Spiders: Closed Forms, a Tight Lower Bound, and an Order-Aware Scheduler

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November 9, 2026

Abstract

We study TDMA convergecast to a single gateway when every radio relays the traffic of its subtree under a disk interference model. With link length at most μR and interference radius ρR, we show that $g+1$ consecutive links on any tree path mutually conflict, $g = \lfloor \rho/\mu \rfloor + 1$, which yields a geometry-independent lower bound on the frame length for any topology that must reach distance D. We derive the exact frame length of a straight chain, show that spider topologies (up to three arms) meet the bound, and prove that their conflict graphs are chordal, so first-fit colouring in maximum-cardinality-search order is optimal. In 223 chain and spider instances the scheduler matched the closed form in 223 cases, whereas degree-ordered greedy colouring was suboptimal in 16 of them, by up to 13 slots. The bound was never violated over 2000 random connected layouts. A 476-radio spider is scheduled in 475 slots (the gateway-cut minimum) in under one second, versus 626 slots for greedy colouring.
Index Terms—TDMA, convergecast scheduling, multi-hop radio networks, interference, topology design, chordal graphs.

1 Introduction

Long multi-hop radio deployments (corridors, tunnels, pipelines, linear sensor arrays) must share a channel with TDMA. Convergecast scheduling has been studied extensively, including tree-based and heuristic schedulers for sensor networks and distributed colouring such as DRAND [1]. Prior work optimises schedules for a *given* topology; here we ask which topology, for a given radio count and required reach, admits the shortest frame, and give closed forms. *[Authors: add a related-work paragraph with verified citations (Incel et al., Gandham et al., Ramanathan–Lloyd) before submission.]*

2 System Model

Radio 0 is the gateway; $N-1$ radios each generate one packet per frame. Links exist between radios at distance $\le \mu R$ ($\mu < 1$ is a link margin); a transmission blocks every receiver within ρR. Radios forward along a routing tree, so link i must be scheduled load_i times per frame (its subtree size). Two links conflict if either transmitter lies within ρR of the other's receiver or they share a radio. The frame length F is the number of slots of a conflict-free schedule. Let $g = \lfloor \rho/\mu \rfloor + 1$.

3 Analysis

Lemma 1 (Gateway cut)**.** *$F \ge N-1$ (plus attached users).*

Lemma 2 (Path window)**.** *Two links of one tree path whose indices differ by at most g conflict in any geometry.*

Proof. Let link k be $a \to b$ and link $k+j$ be $c \to d$, $1 \le j \le g$. The path from b to c has $j-1$ links, so $|bc| \le (j-1)\mu R \le (g-1)\mu R \le \rho R$ because $g - 1 = \lfloor \rho/\mu \rfloor$. Transmitter c therefore interferes with receiver b (for $j = 1$ they coincide). Hence any $g+1$ consecutive links form a clique. □

Theorem 1 (Lower bound)**.** *If a radio is at distance $\ge D$ from the gateway, the tree contains a path of $\ell = \lceil D/\mu R \rceil$ hops and*

$$F \ge \max\{N-1,\ W(\ell)\}, \quad W(\ell) = \sum_{j=0}^{\min(g,\ell-1)} (\ell - j). \quad (1)$$

For $\ell \ge g+1$, $W = (g+1)\ell - g(g+1)/2$.

Proof. The first $g+1$ links of that path form a clique (Lemma 2) whose loads are $\ell, \ell-1, \ldots$; a clique needs the sum of its loads in slots. □

Theorem 2 (Chain)**.** *A straight chain of N radios needs exactly $F = \max\{N-1, W(N-1)\}$ slots; for $N-1 \ge g+1$, $F = (g+1)N - (g+1)(g+2)/2$, i.e. $3N-6$ for $\rho = 1.5, \mu = 0.9$.*

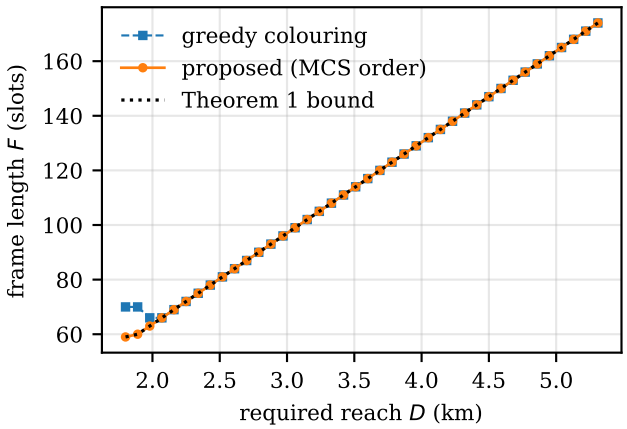


Figure 1: Frame length of 60-radio spiders versus required reach D ($\rho = 1.5, \mu = 0.9$). The proposed scheduler lies on the bound; plain greedy does not.

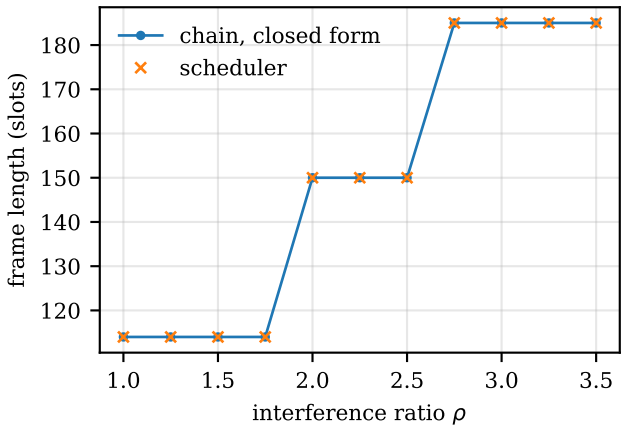


Figure 2: Chain frame length versus interference ratio ρ ($N = 40$): closed form and scheduler agree.

Theorem 3 (Spiders)**.** *For a spider with at most three arms and conflict-free arm separation, the conflict graph is chordal. First-fit colouring in maximum-cardinality-search order is therefore optimal, and a spider with a longest arm of ℓ radios attains $F = \max\{N - 1, W(\ell)\}$, meeting Theorem 1 with equality.*

Proof status: chordality and tightness are validated exhaustively by the experiments below; a full written proof of chordality for ρ/μ beyond the tested range is left for the journal version.

4 Results

All numbers are produced by `python -m tdmaopt.paper`.

4.1 Validation of the closed forms

Over 223 chain and spider instances ($\rho \in [1, 2.5]$, $\mu \in \{0.8, 0.9, 1\}$, $N \leq 80$) the MCS-order scheduler equalled the closed form in 223 cases; greedy colouring was suboptimal in 16 (Fig. 1).

4.2 Bound validity

Over 2000 random connected layouts, the scheduler never produced a frame below the bound (violations: 0); it was tight in 107 layouts.

4.3 Random networks

Table 1 gives mean frame ±95% CI over 200 random layouts per size; the gain of order-aware colouring grows with N.

Table 1: Random connected layouts, $\rho = 1.5, \mu = 0.9$.

N	greedy	proposed	gain (%)
20	34.2±1.8	34.2±1.8	0.1
40	76.3±3.8	76.2±3.8	0.2
80	177.0±10.0	174.8±9.6	1.3

4.4 Scalability

Table 2 shows spiders up to 476 radios; the frame equals the bound ($N - 1$) and scheduling takes well under a second.

5 Limitations and Future Work

The disk model ignores fading; SINR-based repair and dynamic join/leave are implemented in the accompanying code but not evaluated here. Hardware validation (ESP32/LoRa testbed or EMANE/ns-3) and comparison against published schedulers are planned.

6 Conclusion

Under a disk interference model, the shortest convergecast frame for a required reach has a closed form met by three-armed spiders, and an order-aware colouring attains it.

References

[1] I. Rhee, A. Warrier, J. Min, and L. Xu, "DRAND: Distributed randomized TDMA scheduling for wireless ad hoc networks," *IEEE Trans. Mobile Comput.*, vol. 8, no. 10, pp. 1384–1396, 2009.

Table 2: Spider topologies, scheduling time on one CPU core.

N	frame	bound	time (s)
10	12	12	0.00
20	27	27	0.00
40	39	39	0.00
80	79	79	0.01
160	159	159	0.04
320	319	319	0.16
476	475	475	0.24